

3.1.1 Imaging and signalling

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i>	
(i) the formation of a real image by a thin converging lens, understood as the lens changing the curvature of the incident wave-front	HSW9
(ii) the storage of images in a computer as an array of numbers that may be manipulated to enhance the image (vary brightness and contrast, reduce noise, detect edges and use false colour)	Learners are not expected to carry out numerical manipulations in the examination; an understanding of the nature of the processes will be sufficient
(iii) digitising a signal (which may contain noise); advantages and disadvantages of digital signals	HSW12
(iv) evidence of the polarisation of electromagnetic waves.	

Learning outcomes	Additional guidance
(b) <i>Make appropriate use of:</i> (i) the terms: pixel, bit, byte, focal length and power, magnification, resolution, sampling, signal, noise, polarisation <i>by sketching and interpreting:</i> (ii) diagrams of the passage of light through a converging lens (iii) diagrams of wave-forms. (c) <i>Make calculations and estimates involving:</i> (i) the amount of information in an image = no. of pixels $\times$ bits per pixel (ii) power of a converging lens $P = 1/f$, as change of curvature of wave-fronts produced by the lens (iii) use of $\frac{1}{v} = \frac{1}{u} + \frac{1}{f}$ (Cartesian convention) (iv) linear magnification $m = \frac{\text{image height}}{\text{object height}} = \frac{v}{u}$ (v) $v = f\lambda$ including the use of $f = 1/T$ (vi) number of bits, b, provides $N = 2^b$ alternatives; $b = \log_2 N$ (vii) minimum rate of sampling $> 2 \times$ maximum frequency of signal (viii) rate of transmission of digital information = samples per second $\times$ bits per sample (ix) the graphical representation of the digitisation of an analogue signal for a given number of levels of resolution (x) use of $b = \log_2 \frac{V_{\text{total}}}{V_{\text{noise}}}$.	HSW8 using both wave-fronts and rays HSW10 restricted to thin converging lenses and real images; <i>M0.3, M1.1, M2.2, M2.4, M3.2</i> <i>M0.5</i> <i>M2.1</i> <i>M3.1</i>
(d) <i>Demonstrate and apply knowledge and understanding of the following practical activities (HSW4):</i> (i) determination of power or focal length of converging lenses using 3.1.1c(iii) (ii) observing polarising effects using microwaves and light.	links to 3.1.1c(ii)(iii) PAG6 links to 3.1.1a(iv) PAG6